

PERMANENT DENTITION

- **Completion:** The process when all permanent teeth have fully developed and emerged, typically occurring between 18 to 25 years of age, including the third molar.
- **Central Incisor:** The second permanent tooth to emerge in the dental arch, with eruption occurring between the ages of 6 and 7 years.

CHRONOLOGY OF PERMANENT MAXILLARY CENTRAL INCISOR

- **Measurement/Event:** Key milestones in the development of the maxillary central incisor.
- **First Evidence of Calcification:** The initial stage where the tooth begins to form, observed around 3-4 months of age.
- **Crown Completion:** The stage when the crown of the tooth is fully formed, typically completed between 7-8 years of age.
- **Germination:** The process of tooth development when the tooth germ (precursor to the tooth) is present, noted around 10 years of age.
- **Eruption:** The phase when the tooth breaks through the gum line, occurring around 4-5 years of age.

GENERAL FEATURES OF MAXILLARY PERMANENT CENTRAL INCISOR

- **Largest of all incisors:** The maxillary central incisor is the biggest among all incisor teeth.
- **Two in number:** There are two maxillary central incisors, one on each side of the midline of the dental arch.
- **Most prominently seen:** These teeth are the most visible in the mouth.
- **Occludes with lower central and lateral incisors:** They fit together with the corresponding lower central and lateral incisors during chewing.

LABIAL SURFACE

- **Convex:** Curving outward; the labial surface is rounded in both mesiodistal (toward the midline) and cervicoincisally (from the cervical area to the incisal edge) directions.
- **Incisal third:** The upper third of the tooth that is flattened and marked by two faint developmental grooves, dividing it into three mamelons (small rounded projections).
- **Mesial and distal borders:** The edges of the tooth taper downwards toward the cervical area (where the crown meets the root).

- **Distoincisal angle:** The angle at the junction of the distal edge and the incisal edge is rounded, while the mesioincisal angle (the mesial edge meeting the incisal edge) is sharper.
- **Cervical line:** The curve of the gum line that is convex toward the root.
- **Maximum mesiodistal width:** The widest part of the crown is located in the middle third of the tooth.

PALATAL SURFACE

- **Concave:** Curving inward; the palatal surface has a hollow shape with a pronounced convex cingulum (a raised area on the lingual surface).
- **Prominent mesial and distal marginal ridges:** Raised edges that run from the cingulum to the incisal edge, enclosing the lingual fossa (a depression on the lingual surface).
- **Overall surface area:** The total area of the palatal surface is less than that of the labial surface due to the inward slope of the proximal surfaces.
- **Cervical line:** Similar to the labial surface, the cervical line on the palatal side is convex toward the root.

MESIAL AND DISTAL SURFACES

- **Crown obscures the root:** When viewed from the incisal edge, the crown (the visible part of the tooth) covers the root.
- **Labial side:** The front (labial) aspect appears broad and flat in the incisal third.

ROOT

- **Conical:** The shape of the root tapers like a cone.
- **Straight and round on cross-section:** The root is relatively straight and appears circular when cut across.
- **Single root:** The maxillary central incisor typically has one root.
- **Labial, mesiolingual, and distolingual surfaces:** The root has three surfaces: labial (facing the lips), mesiolingual (toward the midline and tongue), and distolingual (away from the midline and tongue).
- **Gradually tapers towards the apex:** The root narrows as it approaches the tip (apex), which is angled distally (toward the back of the mouth).

PULP

- **Spoon-shaped pulp chamber:** The internal space within the tooth that houses the pulp is shaped like a spoon.
- **Two cornua:** Projections within the pulp chamber directed toward the mesial and distal incisal angles.

- **Single root canal:** The pulp chamber contains one canal running down the length of the root.

GENERAL FEATURES

- **Two in number:** There are two maxillary lateral incisors, located between the central incisor and the canine in the upper jaw (maxilla).
- **Second tooth from the midline:** It is the second tooth in the dental arch when counting from the center (midline) outward.
- **Occludes with lower lateral incisor and canine:** This tooth fits together with the corresponding lower lateral incisor and canine during chewing.
- **Resembles the crown form:** The crown (the visible part of the tooth) has a similar shape to the maxillary central incisor but is shorter and narrower.

LABIAL SURFACE

- **Lobulated:** May have rounded projections or lobes on its surface.
- **Slightly convex:** The surface curves outward a little.
- **Distoincisal angle:** The angle at the junction of the distal edge and the incisal edge is more rounded compared to the mesioincisal angle (the mesial edge meeting the incisal edge).
- **Incisal edge slopes mesiodistally:** The biting edge of the tooth tilts from the midline (mesial) towards the distal (back) side.

LINGUAL SURFACE

- **More convex:** The surface curves outward more than the labial surface.
- **Height of contour:** The highest point of the tooth's curvature is located in the middle third of the crown.
- **Cervical line:** The gum line that curves inward toward the root is concave (curving inward) towards the root, especially on the mesial side.
- **Thin incisal edge:** The biting edge is slender and sharp.

MESIAL AND DISTAL SURFACES

- **Wider mesiodistally:** The tooth is broader when measured from the midline to the distal side than from labial (front) to lingual (back).
- **Wedge-shaped crown:** The crown has a tapered or triangular shape.
- **Mesial contour:** The outline of the mesial (inner) surface is convex (curved outward), with the highest point (height of contour) at the junction of the middle and incisal thirds.
- **Mesial surface area:** The area of the mesial surface is larger than that of the distal surface.

INCISAL EDGE

- **Mamelons:** Small rounded projections that may be present on the incisal edge.
- **Labiolingual midpoint:** The biting edge is positioned in the middle when viewed from the front (labial) to the back (lingual).
- **Rounded distal edge:** The edge on the distal side of the tooth is more rounded compared to the mesial edge.

ROOT

- **Slightly longer in proportion:** The root is a bit longer relative to the crown compared to the dimensions of the central incisor.
- **Apex:** The tip of the root, which is inclined (slanted) towards the distal side.
- **Developmental depressions:** Shallow indentations or grooves that are found on the mesial and distal surfaces of the root.
- **More ovoid:** The root has an oval shape compared to the root of the central incisor.

PULP

- **Small pulp chamber:** The internal space within the tooth that contains the pulp is relatively small.
- **Two cornua:** Projections within the pulp chamber that extend toward the mesial and distal incisal angles.
- **Single root canal:** The pulp chamber contains one canal that runs down the length of the root.

MAXILLARY CENTRAL INCISOR

Number: Two (one on each side of midline)

Position: First tooth from the midline

Size: Larger than lateral incisor

Crown Shape: Broad, flat labial surface

Incisal Edge: Straight or slightly curved

Mesioincisal Angle: Sharply defined

Distoincisal Angle: Sharper

Cervical Line: Convex rootwards

Labial Surface: Convex, prominent with two faint grooves

Lingual Surface: Less prominent cingulum, smoother

Lingual Fossa: Less pronounced

Root Length: Shorter relative to crown

Root Shape: Conical, straight

Developmental Features: Two mamelons prominent

Pulp Chamber: Larger pulp chamber with two cornua

Occlusion: Occludes with lower central and lateral incisors

Overall Function: Primary function in cutting food

MAXILLARY LATERAL INCISOR

Number: Two (one on each side of midline)

Position: Second tooth from the midline

Size: Smaller than central incisor

Crown Shape: More narrow and lobulated

Incisal Edge: Slopes mesiodistally

Mesioincisal Angle: More rounded

Distoincisal Angle: More rounded

Cervical Line: Concave rootwards

Labial Surface: May be lobulated and slightly convex

Lingual Surface: More concave with a prominent cingulum

Lingual Fossa: More distinct with a foramen caecum incisivum

Root Length: Longer relative to crown

Root Shape: More ovoid and tapered towards apex

Developmental Features: May have mamelons, thinner incisal edge

Pulp Chamber: Smaller pulp chamber with two cornua

Occlusion: Occludes with lower lateral incisor and canine

Overall Function: Assists in cutting food and aesthetics

DEFINITIONS OF HARD WORDS

Lobulated: Having rounded projections or lobes on the surface.

Cingulum: A convexity on the lingual surface of a tooth, often at the cervical third.

Mesiodistally: Referring to the direction from the midline towards the distal (back) side of the dental arch.

Cervical Line: The line that marks the junction between the crown and the root of a tooth.

Foramen caecum incisivum: An anatomical feature on the lingual fossa of the lateral incisor, sometimes seen as a small pit or depression.

Mamelons: Small, rounded projections found on the incisal edges of newly erupted incisors.

Pulp Chamber: The space within the tooth that contains the dental pulp, which includes nerves and blood vessels.

Occlusion: The way in which opposing teeth come together when the jaws are closed.

Prominent: Standing out or easily noticeable.

Convex: Curving outward.

Concave: Curving inward.

Tapered: Gradually decreasing in thickness or width.

Ovoid: Having an oval shape.

MANDIBULAR PERMANENT CENTRAL INCISOR

General Features

Number: Two in number, located on either side of the midline

Size: Smallest of all permanent teeth

Shape: Long but narrow; crown height > width

Occlusion: Occludes only with maxillary central incisor

Symmetry: Bilaterally symmetric, forming almost a 90° angle with the incisal edge

LABIAL SURFACE

Convex: Curving outward but generally flat in the incisal half

Tapers: Gradually decreases in thickness incisocervically on mesial and distal borders

MESIAL AND DISTAL SURFACES

Profiles: Mirror one another; straight

Height of Contour: At the incisal edge

Crown Shape: Wedge-shaped crown surfaces

Inclination: Inclines more lingually than maxillary incisors

Cervical Line: Concave rootwards, more on mesial aspect

Incisal Border: Straight with sharp mesial and distal incisal angles

INCISAL EDGE

Straight: With three mamelons in newly erupted teeth

Position: Incisal edge is lingual to the labiolingual midpoint

Perpendicular: Edge is perpendicular to a line dividing the tooth labiolingually

LINGUAL SURFACE

Indistinct: Very indistinct lingual anatomy

Surface Area: Overall surface area is less than labial surface

Concave: Except at cervical third, where it is convex due to a less-developed cingulum

Marginal Ridges: Not well-developed; shallow lingual fossa

Cervical Line: Convex rootwards

ROOT

Single-Rooted: Flattened mesiodistally

Grooves: Distal groove is more pronounced than mesial groove

Root Apex: Blunt root apex is distally inclined

Cross-Section: Ovoid in cross-section, wider labiolingually than mesiodistally

PULP

Pulp Chamber: Spoon-shaped pulp chamber with two cornua directed towards mesial and distal angles of the incisal edge

Root Canal: Narrow root canal; rarely has two root canals (buccal and lingual)

DEFINITIONS OF HARD WORDS

Occlude: To come into contact with another tooth during biting or chewing.

Symmetric: Having balanced proportions; equal on both sides.

Convex: Curving outward.

Tapers: To gradually become narrower or thinner.

Inclines: To lean or tilt towards a specific direction.

Concave: Curving inward.

Mamelons: Small, rounded projections on the incisal edges of newly erupted teeth.

Cingulum: A convexity on the lingual surface of a tooth, usually at the cervical third.

Marginal Ridges: Elevated borders of enamel on the mesial and distal surfaces of the tooth.

Fossa: A shallow depression or cavity on a tooth surface.

Ovoid: Having an oval or egg-like shape.

Cornua: Horn-like projections or extensions, typically referring to pulp horns in the pulp chamber.

PERMANENT MANDIBULAR LATERAL INCISOR

General Features

Number: Two in number, second tooth from the midline

Position: Situated between the central incisor and canine

Occlusion: Occludes with maxillary central and lateral incisors

Resemblance: Resembles mandibular permanent central incisor but has key differences

LABIAL SURFACE

Size: Slightly larger than the central incisor

Contour: Mesial contour is straight; height of contour at incisal edge

Incisal Edge: Wider incisal edge compared to central incisor

LINGUAL SURFACE

Indistinct: Indistinct lingual anatomy

Marginal Ridges: Marginal ridges are better developed than in the central incisor

MESIAL AND DISTAL SURFACES

Pronounced: Mesial and distal surfaces are more pronounced

Angles: Distoincisal angle is more rounded than sharp mesioincisal angle

INCISAL EDGE

Position: Incisal edge is lingual to the labiolingual midpoint

Angle: Edge is not at a right angle to a line bisecting the crown

Curvature: Follows the arch curvature, making the distal end more lingually placed

Shape: Incisal edge is curved distally and not straight

ROOT

Length: Root length is greater compared to the central incisor

DEFINITIONS OF HARD WORDS

Occlude: To come into contact with another tooth during biting or chewing.

Contour: The outline or shape of a surface.

Incisal Edge: The cutting edge of an anterior tooth.

Indistinct: Not clearly marked or defined; vague.

Marginal Ridges: Elevated borders of enamel on the mesial and distal surfaces of the tooth.

Pronounced: Strongly marked or clearly defined.

Distoincisal Angle: The angle formed at the junction of the distal and incisal surfaces.

Mesioincisal Angle: The angle formed at the junction of the mesial and incisal surfaces.

Labiolingual Midpoint: The midpoint between the labial (front) and lingual (back) surfaces of the tooth.

Arch Curvature: The natural curve of the dental arch formed by the arrangement of teeth.

Lingually: Pertaining to the tongue or the tongue side of a tooth.

DIFFERENCE BETWEEN MANDIBULAR PERMANENT CENTRAL AND LATERAL INCISORS

Number

- **Central Incisor:** Two (on either side of midline)
- **Lateral Incisor:** Two (second tooth from midline)

Size

- **Central Incisor:** Smallest of all permanent teeth
- **Lateral Incisor:** Slightly larger than the central incisor

Crown Shape

- **Central Incisor:** Long and narrow
- **Lateral Incisor:** Similar shape but wider incisal edge

Symmetry

- **Central Incisor:** Bilaterally symmetric
- **Lateral Incisor:** Less symmetrical compared to central incisor

Incisal Edge

- **Central Incisor:** Straight with three mamelons in new teeth
- **Lateral Incisor:** Curved distally, not straight

Labial Surface

- **Central Incisor:** Convex, generally flat in incisal half
- **Lateral Incisor:** Slightly convex, more lobulated

Mesial Surface

- **Central Incisor:** Straight with height of contour at incisal edge
- **Lateral Incisor:** Straight with a height of contour at incisal edge

Distal Surface

- **Central Incisor:** Straight, forms a right angle with incisal edge
- **Lateral Incisor:** More convex with rounded distofacial angle

Lingual Surface

- **Central Incisor:** Indistinct anatomy, concave
- **Lateral Incisor:** Indistinct anatomy, more convex with cingulum

Root Shape

- **Central Incisor:** Flattened mesiodistally
- **Lateral Incisor:** Slightly longer, more ovoid in cross-section

Pulp Chamber

- **Central Incisor:** Spoon-shaped with two cornua
- **Lateral Incisor:** Smaller pulp chamber with two cornua

DEFINITIONS OF HARD WORDS

Bilaterally Symmetric: Having identical structure on both sides.

Mamelons: Rounded projections on the incisal edges of newly erupted incisors.

Lobulated: Having lobes or rounded projections.

Convex: Curved outward.

Concave: Curved inward.

Cingulum: A rounded ridge at the base of the crown on the lingual surface of anterior teeth.

Ovoid: Egg-shaped; rounded.

Cross-section: A cut through an object to reveal its internal structure.

Cornua: Horn-like projections or extensions.

Height of Contour: The highest point of the contour on a tooth surface.

LINE ANGLES AND POINT ANGLES IN MAXILLARY CENTRAL INCISOR

Line Angles of the Maxillary Central Incisor

- **Mesiolingual:** Angle between mesial and lingual surfaces
- **Distolingual:** Angle between distal and lingual surfaces
- **Mesiolabial:** Angle between mesial and labial surfaces
- **Linguoincisal:** Angle between lingual and incisal surfaces
- **Labioincisal:** Angle between labial and incisal surfaces

Point Angles of the Maxillary Central Incisor

- **Mesiolabioincisal:** Angle formed by mesial, labial, and incisal surfaces
- **Distolabioincisal:** Angle formed by distal, labial, and incisal surfaces
- **Mesiolinguoincisal:** Angle formed by mesial, lingual, and incisal surfaces
- **Distolinguoincisal:** Angle formed by distal, lingual, and incisal surfaces

DEFINITIONS OF HARD WORDS

Line Angle: The line formed by the intersection of two tooth surfaces.

Point Angle: The point formed by the intersection of three surfaces of a tooth.

Mesiolingual: Referring to the mesial (toward the midline) and lingual (toward the tongue) surfaces.

Distolingual: Referring to the distal (away from the midline) and lingual surfaces.

Mesiolabial: Referring to the mesial and labial (toward the lips) surfaces.

Linguoincisal: Referring to the lingual and incisal (cutting edge) surfaces.

Labioincisal: Referring to the labial and incisal surfaces.

Mesial: The surface of a tooth that is closest to the midline of the dental arch.

Distal: The surface of a tooth that is farthest from the midline of the dental arch.

Incisal: Referring to the cutting edge of anterior teeth.

Cervical: Relating to the neck or lower part of the tooth, near the gum line.

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